

Partial Denture Base

Learning Objectives:

- Understand the functions of the denture base.
- Identify the different types of denture bases according to the material.
- Discuss the advantages and disadvantages of acrylic resin and metal denture bases.
- Recognise the different types of artificial teeth.
- Explain the attachment of artificial teeth to the denture base.

Partial Denture Base

It is that part of the partial denture that supports the artificial teeth, rests on the foundation tissues, and transfers the occlusal force to the supporting structures.

Functions of the denture base

1. **Masticatory function:** It supports the artificial teeth and consequently receives the functional forces from occlusion and transfers functional forces to supporting oral structures.
2. **Cosmetic function:** This is related to reproducing natural-looking contours.
3. **Physiological function:** The stimulation of the underlying tissue of the residual ridge by massaging action during movement of the denture base under functional stress (within physiological tolerance) maintains its form and tone better than similar tissues suffering from disuse (disuse atrophy).
4. **Prevent vertical and horizontal migration of remaining natural teeth.**
5. **Secondary retention** is provided by the intimate relationship of denture bases with the underlying tissue



Types of Denture Base according to the material:

A. Acrylic Resin Denture Base

It is the most widely used type of denture base because of its easy fabrication and attachment to a metal framework.

Sufficient thickness of resin (1.5 mm) should be placed under the gridwork minor connector. Thickness is necessary to prevent the weakness and fracture of the acrylic resin base material surrounding the metal framework.

Advantages:

1. Can be easily relined.
2. Easy to fabricate, adjust, finish, polish, and repair.
3. Resin is more aesthetic than metal.

Disadvantages:

1. More porous than metal and therefore more difficult to clean.
2. Requires more bulk for rigidity than metal.
3. Acrylic is a poor thermal conductor.

B. Metal Denture Base

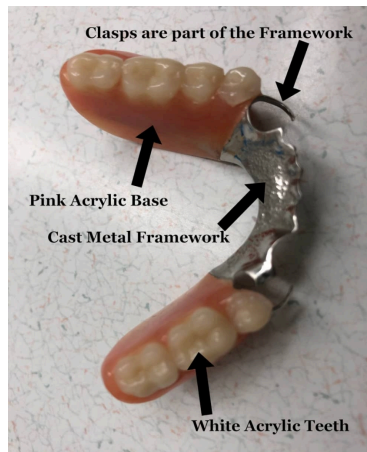
The denture base can be made of gold, platinum, or most commonly cobalt-chromium. Such a type can be used in cases of a short span with no aesthetic concerns. It is also indicated when there is not enough inter-ridge space for artificial teeth (inadequate intermaxillary space) because of over-eruption of opposing teeth.

Advantages:

1. The inherent cleanliness of the cast metal base contributes to the health of oral tissue when compared with an acrylic resin base.
2. The thermal conductivity of metal contributes much to the patient's acceptance of a denture and may help prevent the feeling of the presence of a foreign body.

Disadvantages:

1. Difficult to reline, adjust, and rebase.
2. Expensive.
3. Poor esthetic



Types of Artificial Teeth

- Acrylic teeth: have the ability to be attached chemically to the denture base.
- Porcelain teeth: attached to the denture base by mechanical means.
- Metal teeth: In some cases, the molar teeth may be processed as part of the denture base by a casting procedure. This is indicated in cases of limited inter-maxillary spaces.

